

M. PHARM
(SEM I) THEORY EXAMINATION 2022-23
CELLULAR AND MOLECULAR PHARMACOLOGY

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 2 = 20**

- (a) Enlist the functions of mitochondria.
- (b) Interpret the importance of siRNA and micro-RNA.
- (c) Give the clinical significance of NO.
- (d) Review the pathway of MAPK signaling.
- (e) Review the terms proteomics and transcriptomics.
- (f) What are micro-array techniques?
- (g) Elaborate the applications of metabolomics.
- (h) Elaborate the clinical applications of immunotherapeutics.
- (i) Compare primary and secondary cell culture.
- (j) Define biosimilars.

SECTION B

2. Attempt any two parts of the following:**2 x 10 = 20**

- (a) Explain the intrinsic and extrinsic pathways of apoptosis.
- (b) Summarize the role of secondary messengers in cell signalling pathway.
- (c) Document the principles and methodology of cell viability assays.

SECTION C

3. Attempt any five parts of the following:**5 x 7 = 35**

- (a) Explain the pathways involved in cell cycle and what are the regulators of cell cycle?
- (b) Interpret the mechanism of GPCR (adenylyl cyclase and IP₃-DAG) mediated physiological actions.
- (c) Differentiate between autophagy, necrosis, and apoptosis.
- (d) Examine the various types of gene transfer techniques.
- (e) Explain the concept of ELISA with diagrammatic representation.
- (f) Discuss the various vectors used in rDNA technology.
- (g) Describe the cryopreservation techniques used in cell culture.